

ADR 0105 — Share the measurement, not the floor

Status: Accepted — implemented **Date:** 2026-09-02 **Issues:** [#581](#) — the documented git floor is 2.32, but `revert_would_conflict` has silently required 2.38 since [#327](#) **Supersedes:** nothing · **Superseded by:** nothing **Extends:** [ADR 0099](#), which introduced the per-feature version gate for the graph preview

Context

`docs/SUPPORTED_VERSIONS.md` declares a **product** floor of git **2.32**, derived from `GIT_CONFIG_GLOBAL`. Since [#327](#), `activity::revert_would_conflict` has run:

```
git merge-tree --write-tree --merge-base=<commit> <head> <parent>
```

`--write-tree` arrived in git **2.38**. There was no version check of any kind on that path — the function read git's exit code and mapped `0` to "clean", `1` to "conflict", and everything else to an unexplained `Err`.

Measured 2026-09-02, running that exact argv against two real gits in rootless podman containers:

git	exit	output
2.34.1 (Ubuntu 22.04 LTS)	129	usage: git merge-tree <base-tree> <branch1> <branch2>
2.43.0 (Ubuntu 24.04 LTS)	0	the merged tree oid

129 is neither 0 nor 1. So on any host in the documented 2.32–2.37 band the revert offer **silently never appeared**, and the user was told nothing.

The posture was never unsafe. `revert_offer_established` treats `Err` exactly like a conflict, and its own comment says *"'couldn't tell' must never read as 'safe to offer'"*. That is why this survived three years and several audits: it degrades fail-closed, and a fail-closed degrade produces no symptom anyone reports. What was missing was not safety. It was the **explanation**.

**User opens the
undo menu
on a host with git
2.34.1**

```
graph TD; A["User opens the undo menu on a host with git 2.34.1"] --> B["revert_would_conflict"]; B --> C["git merge-tree --write-tree ..."]; C --> D["exit 129 usage: git merge-tree ..."];
```

`revert_would_conflict`

`git merge-tree --
write-tree ...`

exit 129
usage: git merge-
tree ...

```
graph TD; A[ ] --> B[Err(String) — the 'no answer' arm]; B --> C[revert_offer_established → false]; C --> D[The Revert entry is simply absent. No error, no reason, no clue];
```

Err(String) — the
'no answer' arm

revert_offer_established
→ false

**The Revert
entry is simply
absent.**

No error, no
reason, no clue

Meanwhile #576 had already built exactly the machinery this needed — a once-per-process probe, a parser for git's `--version` line, and a pure

comparison — and it was **private to** `preview.rs`. #581's own words: *"That gate is worth building once, for both callers, rather than twice."*

Decision

Share the measurement. Do not share the floor.

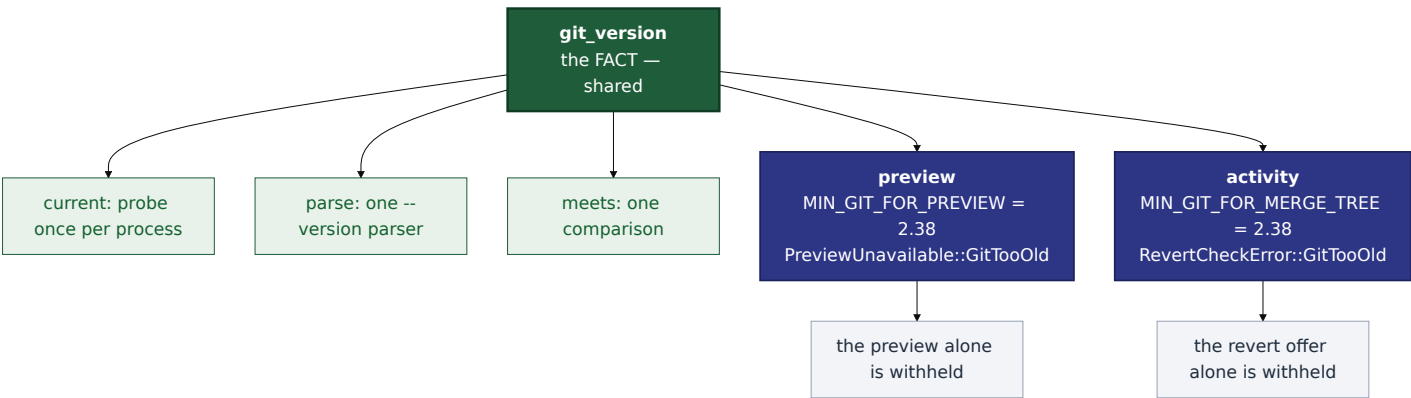
A new module, `crates/git-vista-server/src/git_version.rs`, owns the *fact*:

- `current(repo)` — the probe, cached per process in a `OnceCell`, caching only successes so a transient failure does not permanently disable a feature;
- `parse(line)` — one parser for git's `--version` line, including the vendor suffixes (`2.39.5 (Apple Git-154)`, `2.43.0.windows.1`);
- `meets(found, floor)` — one pure comparison, on `(major, minor)`.

Each feature keeps its **own** floor constant and its own refusal type:

Feature	Constant	Refusal
Graph preview	<code>preview::MIN_GIT_FOR_PREVIEW</code>	<code>PreviewUnavailable::GitTooOld { found, minimum }</code>
Revert offer	<code>activity::MIN_GIT_FOR_MERGE_TREE</code>	<code>RevertCheckError::GitTooOld { found, minimum }</code>

Both are `(2, 38)` today. That is a **coincidence of the same plumbing, not a shared policy**, and the ADR's title is the whole point: a single shared constant would read as authoritative, and the next feature needing 2.41 would either raise it for everyone or quietly fork it back. A shared *number* becomes a second product floor by the back door. A shared *measurement* is what was actually missing.



revert_would_conflict's error becomes two facts, not one.

`RevertCheckError::GitTooOld { found, minimum }` is a property of the host that will be true again next time; `RevertCheckError::CheckFailed(String)` may not be. The Recovery Center carries the distinction through as `CheckFailedReason::GitTooOld`.

The safety posture does not change. Both arms still decline to offer. `revert_offer_established's .unwrap_or(false)` is untouched. The split buys an explanation, never a different answer.

This stays out of the boot gate, for the reason ADR 0029 already gives: `sandbox::probe` is deliberately the one gate in this codebase with no degraded outcome, and a *capability* question does not belong in a gate whose whole argument is that it has none.

Alternatives considered

Raise the product floor to 2.38. Rejected. A host on 2.32–2.37 is a fully supported host on which every other feature is correct; refusing it service for one menu entry is a much larger harm than withholding that entry. It would also have to be enforced somewhere, and the only boot gate available is the one that must not grow a degraded mode.

One shared `MIN_GIT` constant. Rejected — the title of this ADR. The two floors coincide today for one reason (`merge-tree --write-tree`), and reading a call site should tell you what *that feature* needs without chasing a table.

Leave the error a single `Err(String)` and just document the floor. Rejected. It would fix `SUPPORTED_VERSIONS.md` and leave the user on 2.34 watching a menu entry not appear. The distinction is cheap and the whole cost of the defect was that the two cases were indistinguishable.

Put `GitTooOld's` version numbers on `CheckFailedReason`. Rejected. `CheckFailedReason` is `Copy` and caller-facing; a host detail that changes under it belongs in `RevertCheckError`, which is where it lives.

Consequences

- One place now establishes the running git's version, which is what #581 asked for. A third feature needing a floor adds a constant, not a mechanism.
- A user on git 2.32-2.37 is told *why* the revert offer is missing.
- The honest limit, stated rather than hidden: the version is probed **once per process**, so an operator who upgrades git under a running server does not get the gated features until restart — the same posture `sandbox::capabilities::current()` already takes toward host capability.
- `revert_would_conflict` now costs one extra `git --version` per *process*, not per menu-open.
- **A testing consequence worth naming**, because it nearly produced a vacuous test. `git_version::current` caches per process and cannot be made to lie, so no test can make this host's git look old — and this project's hosts are all far above 2.38. A pure gate test alone would have proven the decision correct while leaving *deletion of the gate call* completely green. `revert_would_conflict_at_version` therefore takes the version rather than probing it; everything else on that path is production. Mutation-proved two ways, failing differently:

Mutation	Kind	Result
delete the <code>if let Some(too_old)</code> gate call	removes the mechanism	caught — only the wiring test goes red
<code>MIN_GIT_FOR_MERGE_TREE</code> 2.38 → 2.32	weakens the mechanism	caught — the pure gate test <i>and</i> the wiring test go red, on different assertions

- `docs/SUPPORTED_VERSIONS.md`'s feature-floor table gains the row that was true from #327 and undocumented until now.